

جامعة بيروت العربية

BEIRUT ARAB UNIVERSITY

NONLINEAR HOUSE PRICE MODELING

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CMPS470 – Deep Learning

PROJECT OVERVIEW

1

Predict house prices using deep learning

2

Compare shallow and deep neural networks

3

Study nonlinear relationships between features

4

Analyse hidden representations using PCA

DATASET

- 2930 residential properties
- 79 features
- Numerical and categorical data
- Contains missing values
- Dataset taken from Kaggle

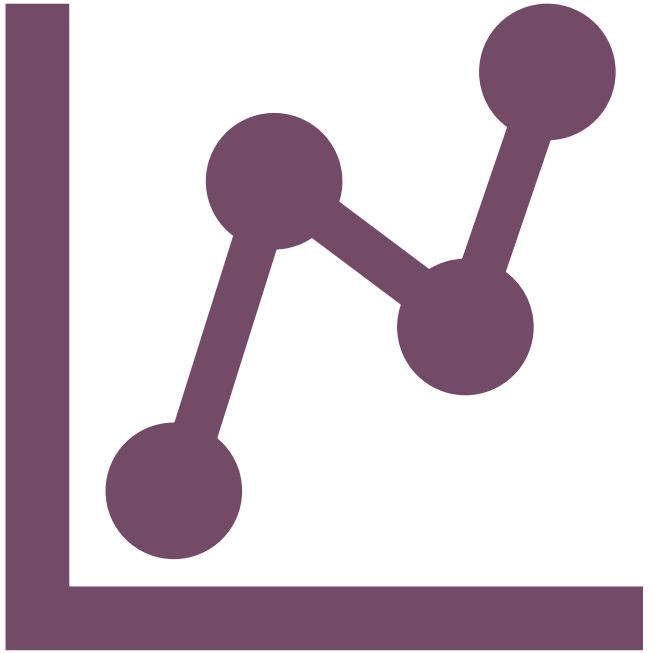
DATA PREPROCESSING

- Filled numerical missing values using median
- Filled categorical missing values with “None”
- Applied one-hot encoding
- Applied log transformation
- Created:

✓ TotalArea

✓ HouseAge

✓ GarageAge



MODELS USED

Model	Purpose
Linear Regression	Baseline
1-layer Neural Network	Simple nonlinear learning
3-layer Neural Network	Medium depth
5-layer Neural Network	Deep learning model

RAW PRICE VS LOG PRICE

- Compared raw SalePrice and log(SalePrice)
- Used same architecture for fair comparison
- Log transformation reduced large prediction errors
- Residuals became more stable

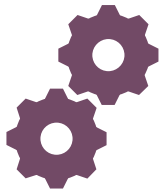
RESULTS & COMPARISON

Model	Observation
Linear Regression	Highest error
1-layer NN	Better performance
3-layer NN	Best overall
5-layer NN	Slight overfitting

PCA VISUALIZATION



Extracted hidden layer activations



Applied PCA to reduce dimensions



Visualized hidden feature clusters



Some neighbourhoods formed meaningful groups

CONCLUSION

- Deep learning improved prediction performance
- Feature engineering was very important
- Log transformation improved stability
- Increasing depth improved learning up to a point
- Very deep models may overfit